3. DESIGN APPROACH

The Design Requirement Specifications section defines what Forest Buddy accomplished through its system requirements and subsystem functions. This section explains the design options, overall system operation, and how the subsystems work together to meet those requirements. Forest Buddy uses and integrated hardware and software system to travel through forested terrain while collecting GNSS positioning and signal measurements. The camera vision subsystem tracks the operator during the initial trip, while the navigation subsystem records waypoints along the route. During operation, GNSS positioning and IMU orientation data guide the rover between waypoints, and LiDAR checks for obstacles along the travel path. The locomotion subsystem controls the motors to move and steer the rover. The communication and data processing subsystems collect, combine, and transfer measurements from the rover to the base station. The data analysis subsystem displays the results through the website maps and graphs. Component selections account for operating time, outdoor conditions, and two-semester development timeline.

3.1. Design Options

Two designs were considered for communication between Forest Buddy and its base station. The first uses a drone as an elevated radio relay, while the second uses a rover that is semi-autonomous with manual rover control available if connectivity is lost. The drone offers a possible communication path above the trees and terrain but adds equipment cost, flight coordination, and batter limitations. The semi-autonomous design reduces cost and complexity, although obstacles can interrupt communication and require operator assistance. Both options are evaluated against the project's communication requirements, budget, operating time, and development timeline. Design option 2 is selected because it leaves more of the budget available for the rover's sensing and navigation hardware.

3.1.1. Design Option 1

Design Option 1 uses a drone hovering above the tree line to receive and retransmit radio data between the drone and base station. Elevating the relay can improve the radio path when trees or terrain obstruct direct communication. The drone carries telemetry and Real Time Kinematic (RTK) data and uses GNSS satellite signals. This approach could extend communication coverage, but the drone, relay hardware, and batteries increase cost. Flight time also limits relay availability; this option is less favorable because it uses a larger share of the project budget.

3.1.2. Design Option 2

Design Option 2 omits the drone and instead requires the operator to manually drive the rover if the rover loses connectivity. We chose this option because a drone would eat too much of the project's budget. This design option uses a Jetson Orin Nano as its primary controller. The Jetson allows for an easier way to implement an AI model to track a human through a camera. The rover detects and tracks a human to follow through the forest while the user sets waypoints for the rover to follow. The rover only uses human detection for the first path through the forest. The controller has the option to control the rover manually if Forest Buddy cannot travel around an object. This design also uses two GNSS antennas from SparkFun, two ZED-F9P RTK boards and two telemetry radios. The GNSS antennas collect satellite data overhead while the boards help with RTK positional accuracy. The accuracy of these boards will be within 1 cm of where the rover is. The telemetry radios send data between the base station and rover.

3.2. System Overview

Forest Buddy consists of six connected subsystems: locomotion, navigation, camera vision, data processing, communication, and data analysis. The onboard controller coordinates sensor information and movement requests so the rover can follow the operator, record waypoints, and later travel between the waypoints. The locomotion subsystem uses four motor driven wheels and motor drivers to control movement and steering. Camera vision provides operator tracking during initial route recording, while navigation combines GNSS positioning data, IMU orientation, and LiDAR distance measurements to guide the waypoint path. If an obstacle blocks the path, the rover stops and returns to the previous waypoint to wait for the operator to clear the path.

The communication subsystem supports GNSS positioning, communication between the rover and base station, and manual remote control. The data processing subsystem combines GNSS and environmental measurements into records for storage and transmission. The data analysis subsystem uses these records to display the rovers path and signal attenuation through maps and graphs on the Forest Buddy website.

At Level 0, Forest Buddy is represented as one system that receives batter power, operator input, satellite signals, and information from the surrounding environment. Its main outputs are rover movement, recorded measurements, and displayed results. The Level 1 diagram expands this view by showing how the six subsystems exchange information and commands. Together, these subsystems provide a platform for collecting and reviewing GNSS signal measurements along forest routes.



Fig. 3-1: Forest Buddy System at a Glance (Level 0)

Figure 3-2 shows a more detailed view of Forest Buddy. It visualizes how the inputs and outputs of the different components of each subsystem will interact with the microcontroller.

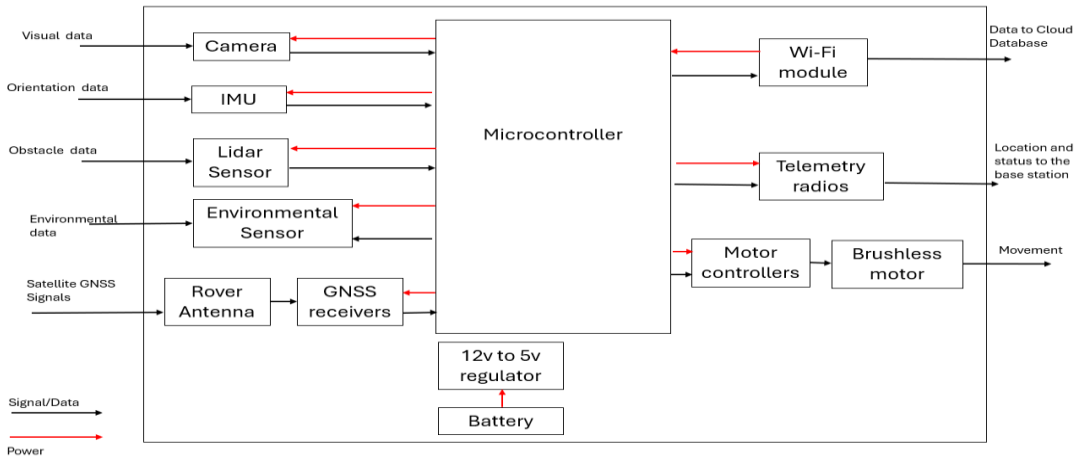


Fig. 3-2: Forest Buddy Functionality (Level 1)

3.2.1. Microcontroller

Multiple microcontrollers were evaluated to determine which option would best support Forest Buddy's processing and Ai requirements. Table 3-1 compares the NVIDIA Jetson Orin Nano, Raspberry Pi 5, and Orange Pi 5 Plus on cost, RAM, power consumption, available interfaces, and AI computing capabilities. These were considered because we needed a microcontroller that can handle processing camera data, YOLOv8 human detection, and communicating with a Pixhawk 4 to control the navigation and movement.

Table 3-1: Micro Controller Options

Product	Cost	RAM	Power	Interfaces	Ai Compute
Requirements	≤$150		≥ 6		
Jetson Orin Nano[39]	$400	8 GB LPDDR5	7 to 25 W	40-pin header, 4 USB ports, 2 camera ports	67 TOPS
Raspberry Pi 5 (8GB)[40]	$180	8 GB LPDDR4X	5 to 12 W	40-pin header, 4USB ports, 2 camera ports	None built in
Orange Pi 5 Plus (8GB)[41]	$225	8 GB LPDDR4X	5 to 10 W	26-pin header, 3 USB ports, 3 camera ports	6 TOPS

The NVIDIA Jetson Orin Nano (shown in Fig 3-3) was selected because of its AI processing capabilities, memory, and compatibility with the software Forest Buddy uses. It provides 8 GB of LPDDR5 Ram and up to 67 TOPS of AI computing performance. This allows for YOLOv8 to be implemented for human detection and tracking. The Raspberry Pi 5 and Orange Pi 5 Plus are far less expensive, but the Jetson offered much more AI computing capabilities. The team can also get access to NVIDIAS software tools for the computer vision applications. The Jetson also comes with many ports, camera ports and 40 pins to connect hardware.



Figure 3-3 NVIDIA Jetson Orin Nano[39]

The Jetson Orin Nano will handle human detection using YOLOv8 and will also help with the rovers navigation and movement by communicating with Pixhawk 4.

3.3. Subsystem Overview

Forest Buddy consists of six subsystems that work together to make a semi-autonomous forest ranging rover:

1) The locomotion subsystem is the chassis of the rover while also allowing Forest Buddy to move around.

2) The navigation subsystem is what helps the rover maneuver forested terrain. It prevents the rover from harming itself, the user, or forested land through different sensors to tell its distance from objects and different algorithms to follow a set path safely.
3) The camera vision subsystem involves human tracking, so Forest Buddy can follow the user through its first path in the woods before driving autonomously.
4) The data processing subsystem gets environmental data and packets GNSS data.
5) The communication subsystem collects GNSS data to help with rover positioning and collects attenuation through the trees to tell how healthy the forest is while also including a remote controller to manually operate Forest Buddy if it is needed.

3.3.1. Locomotion

The data analysis subsystem takes the data from the communication subsystem and data processing to provide heat maps of healthy areas in forests and manages a website that holds this data for review. For locomotion, the rover will use four wheels, a motor attached to each wheel, and two motor drivers. The motor drivers will be wired directly up to both wheels on the left side and the right side and will turn the rover by spinning one side faster than the other. Listed in Table 3-1 are possible motor options.

Table 3-2: Motor Options

Motor	Requirement	90 watt 60D Planetary Gear Motor [1]	70:1 Metal Gearmotor 37Dx54L [2]	Thor 2318 Planetary Gear Brushless Motor [3]
Image				
Cost	<$250	(1) $123.50	(4) $143.80	(4) $239.96
Input Voltage	12V	24V	12V	12V
Rated Torque and Speed	1.5-3.0 Nm	9.7 Nm @ 89 Rpm	2.7 Nm @ 150 Rpm	2.7 Nm @ 215 Rpm
Brushless	Optional	No	No	Yes

Four Thor 2318 Planetary Gear Brushless Motors (shown in Table. 3-2) are needed to turn all four wheels. This motor provides similar torque and greater speed than the other options. Being brushless is a bonus but was not the deciding factor. The 60D was not chosen because it is too expensive to be used with all four wheels, and designing a transmission system for one motor would add too much complexity for a two-semester project. The 37Dx54L, while much cheaper than the Thor, is too slow for the project.

Each Thor 2318 uses approximately 60 watts at full power. To satisfy the project's one-hour operating time requirement, a 25-amp hour battery will be used. 60 * 4 * 1 hour * 100% active time = 240 watt hours, 240 / 12 V = 20-amp hours. The other components use comparatively little power, so the extra 5 amp-hours of battery will provide plenty of headroom.

These motors need a motor driver to function. Listed in Table 3-3 are possible motor driver options.

Table 3-3: Motor Driver Options

Motor Driver	Requirement	Bidirectional 20A Electronic Speed Controller [4]	70A Bidirectional ESC AM32 Brushless Controller, 2-6S, XT60 [5]	Bidirectional 45A Electric Speed Controller, 2-6S Lipo ESC with XT60 [6]
Image				
Cost	<$20	$18.99	$49.99	$28.99
Input Voltage	12V	12V	12V	12V
Max Amps	>16A	20A	70A	45A

Two Drivers are needed, one for each side, to turn the rover. The Bidirectional 20A Electronic Speed (shown in Table. 3-3) is the cheapest option that meets all requirements.

3.3.2. Navigation

The navigation subsystem helps Forest Buddy follow a path through the forest while detecting obstacles. It combines LiDAR, an inertial measurement unit (IMU), and a navigation algorithm to guide the rover. The subsystem also uses GNSS position data from the communication subsystem and sends movement commands to the locomotion subsystem.

Table: 3-4 Lidar Options

Product	Cost	Detection Range	Accuracy	Scanning Coverage	Communication Interface
Requirements*	≤ $150 per sensor	Detect obstacles at 5 ft	Error within ±0.20 ft at 5 ft	Forward detection; wider coverage preferred	UART or I^2C
FHL-LD19 Lidar Sensor[7]	$71.90	0.07–39.4 ft	±0.15 ft average (0.98–39.4 ft)	360° scanning	UART USB adapter
TFMini-S - Micro LiDAR Module[8]	$72.95	0.33–39.4 ft	±0.20 ft@ (0.33–19.7 ft); ±1%@ (19.7–39.4 ft)	Single point Approx. 2° FOV	UART / I^2C
Garmin LIDAR-Lite v4 LED[9]	$119.95	0.16–32.8 ft	±0.033 ft up to 6.56 ft; ±0.066 ft up to 13.1 ft; ±0.164 ft up to 32.8 ft	Single direction 4.77° beam divergence	I^2C / ANT

The FHL-LD19, which can be seen in Figure 3-4 is selected because it provides 360-degree scanning at a lower listed cost than the other options. Its scanning coverage allows Forest Buddy to measure distances in multiple directions, which supports obstacle detection and path planning. The TFMini-S and Garmin LIDAR-Lite v4 measure distance in a single direction and require additional sensors or a scanning mechanism to provide similar coverage.



Figure 3-4 FHL-LD19 Lidar Sensor

The IMU measures acceleration and angular velocity to help estimate the rover's orientation as it moves. This information helps track turns and changes in tilt while traveling over uneven ground. The IMU works alongside the GNSS position data to support navigation along the intended path.

Table 3-5: IMU Options

Product	Cost	Functionality	Orientation Accuracy	Orientation Processing	Communication Interface
Requirements	≤ $50 per board	9DoF Accelerometer, gyroscope, magnetometer	Orientation error ≤ 5° in rover tests	Support rotation; onboard fusion preferred	I²C, SPI, or UART
SparkFun VR IMU Breakout - BNO086 (Qwiic)[10]	$34.95	9DoF Accelerometer, gyroscope, magnetometer	Rotation vector: 2.0° static error; 3.5° dynamic error	Onboard sensor fusion; rotation vector output	I²C, SPI, UART
SparkFun 6DoF IMU Breakout - BMI270 (Qwiic)[11]	$18.50	6DoF Accelerometer and gyroscope	Fused orientation accuracy not specified on product page	External processing needed for orientation; no onboard magnetometer	I²C, SPI
SparkFun 9DoF IMU Breakout - IM19[12]	$199.95	9DoF GNSS/IMU fusion module	Roll/pitch: ±0.025°; heading: ±0.25°	Onboard GNSS/IMU fusion; requires RTK fixed NMEA input and PPS timing	3 UARTs; USB-to-UART on board

SparkFun BNO086, which can be seen in Figure 3-5 is selected because it has 9 degrees of freedom(DOF).



Figure 3-5: SparkFun BNO086 IMU

The SparkFun BNO086 combines an accelerometer, gyroscope, and magnetometer with onboard sensor fusion to provide orientation estimates. The BMI270 requires additional orientation processing, while the IM19 is more expensive.

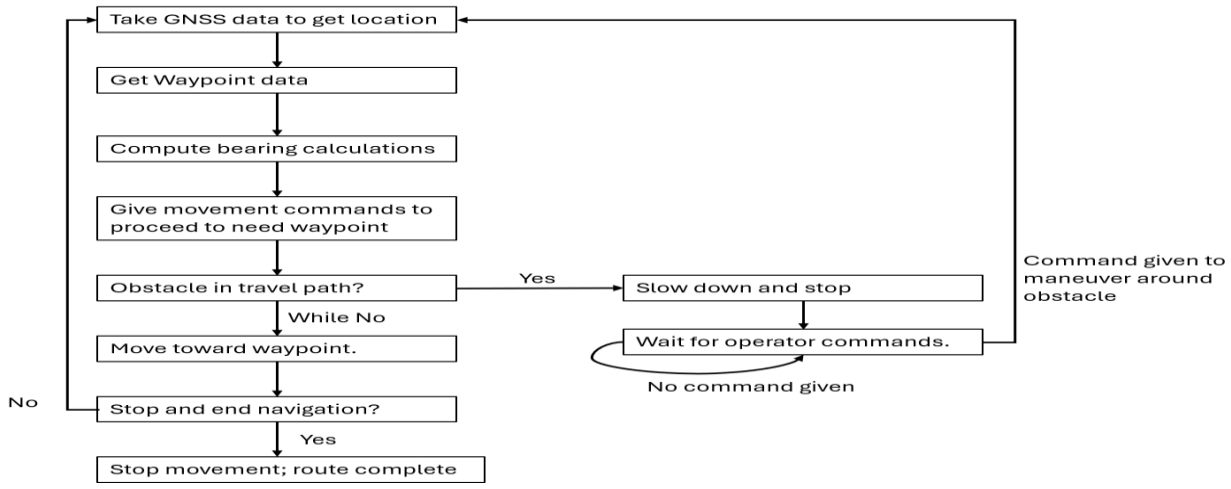


Figure 3-6: Navigation Algorithm

Figure 3-6 shows how the navigation algorithm uses GNSS data to determine the rover's current location and compares it with the waypoint data to calculate the bearing toward the next waypoint. The algorithm then sends movement commands to the locomotion subsystem to guide the rover along the path. During travel, the LiDAR checks for obstacles in the rover's direction of movement. If an obstacle blocks the path, the rover stops, returns to the previous waypoint, and waits for the operator to clear the path. If no obstacle is detected, the rover continues toward the waypoint and stops once navigation is complete.

3.3.3. Camera Vision

The camera vision subsystem handles detection and tracking of the operator. This allows Forest Buddy to establish a safe path through the forest by setting waypoints behind the operator. The system will keep the them in frame and calculate the operator's distance from the rover by using camera geometry, this will be used to determine what movement inputs need to process to keep a safe following distance. The list of possible camera options can be seen in table 3-6.

Table: 3-6 Camera options

Microcontroller	Cost	FOV	Power Consumption	Connection Type	Distortion due to Lense	FPS
Logitech Webcam C920 [13]	$67.25	78°	2.5 watts	USB	Low	30
IMX219 8MP CSI Camera Module [14]	$21.90	120°	Less than 1 watt	CSI	Mild	30
Arducam IMX219 Visible Light Fixed Focus Camera Module [15]	$12.08	65°	Less than 1 watt	CSI	Low	30

Figure 3-7 shows the Logitech Webcam C920, it was chosen because of its balance between FOV and lens distortion.



Figure 3-7: Logitech Webcam C920

Its 78-degree field of view keeps the person in frame enough without the distortion that would be caused by the 120-degree fisheye lens, which would make distance estimation less accurate. The 65-degree FOV from the Aruducam is too narrow and is likely to lose track of the operator if they move sharply. The C920's low distortion means that the calibration will be simple, and its calibration steps are well documented. The C920 is also a very common USB camera option with the Jetson Orin Nano, so its integration guides and support are easy to find. Although the cost and power draw is the highest, its pros of the FOV to distortion ratio, ease of integration, as well as the documentation for it outweighs the cons.

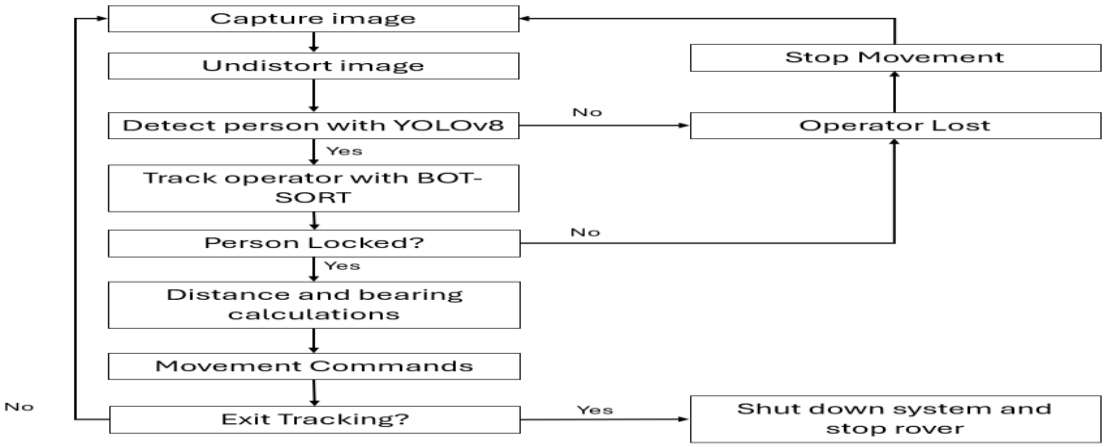


Figure 3-8: Computer Vision Software

The above figure 3-8 visualizes the functionality of the Vision software, it starts by capturing frames from the camera, undistorting it, then detecting the operator by using YOLOv8, tracks them with BOT-SORT, and using the tracking box it calculates the distance of the operator from the rover which will be used for movement commands. The software begins by reading frames from the C920 and using OpenCV to undistort it and then using YOLOv8 to detect the operator in view, which it then keeps only the "person" class above a certain confidence threshold. The detection then goes through BOT-SORT, which gives the person an ID allowing detection frames to be linked together. Because it can predict a person's movement between frames, the ID can survive if they are shortly obstructed by a tree or obstacle. The software then estimates the operator's distance and bearing, it then compares the distance with the target following gap and sets the speed.

3.3.4. Data Processing

The Data Processing subsystem handles collecting temperature, humidity, pressure and luminosity data, using sensor fusion to incorporate this data with the receiver data, and packeting timestamping this data for communications to transmit. For the temperature and pressure recording done on the rover, an environmental sensor has been chosen. The data from this incorporated sensor is compared with the GNSS recorded data. The sensor's data is combined with GNSS data and packeted before transmission or local storage, and must be compatible with both the Jetson Nano microprocessor, and Ardupilot. The temperature sensor board, and the luminosity photoresist are connected to the Listed in Table 3-7 are possible sensor options.

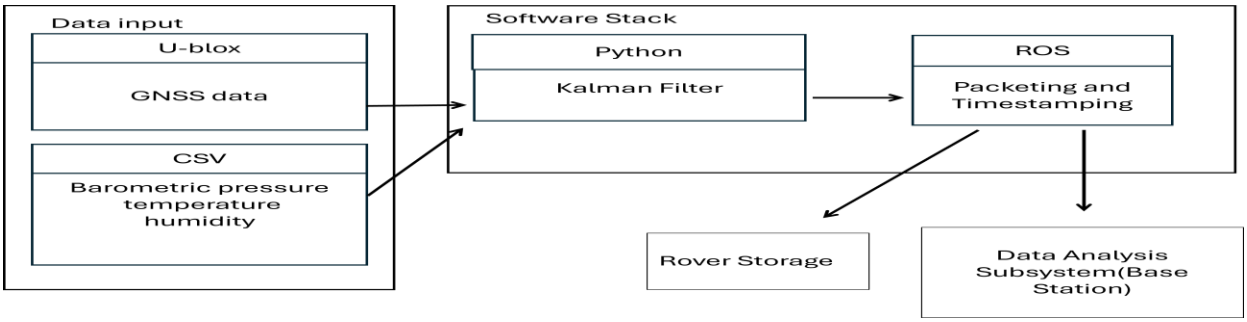
Sensor	Cost	Functionality	Accuracy	Connectivity
Requirements	<$30	Barometric Pressure, temperature, humidity	<±100Pa,±1.0°C,±0.1%RH	I2C
ENS160/BME280[16]	$39.95	Barometric Pressure, temperature, humidity, gas, altitude	±12Pa,±0.5°C, ±0.008%RH	I2C, UART
SHT31[17]	$14.95	Barometric Pressure, temperature, humidity	1 hPa,±1°C, ±3% RH	I2C
BME688[18]	$24.95	Barometric Pressure, temperature, humidity, gas	±60Pa,±1.0°C,±0.008%RH	I2C

Figure 3-9 shows the SparkFun Environmental Sensor – BME688, which was chosen for its cost and accuracy.



Fig. 3-9: SparkFun Environmental Sensor - BME688

Shown in Figure 3-9 is the Sparkfun BME688 environmental sensor. This sensor is under the cost requirement, features required connectivity and features sensors for measuring barometric pressure, temperature and humidity as required, alongside VOC gas. The BME688 does not outperform the combo ENS160/BME280 with accuracy but is still within the required accuracy and is in the required budget.



The software flowchart depicted in Figure 3-10 shows how the input data from the rover's GNSS receiver and environmental sensor breakout board are combined through a Kalman filter. These parameters create redundancy with sensor measurement, altitude from both GNSS and the pressure sensor, and humidity measures from both the sensor and estimates from the receiver data. After the data is packeted and timestamped, most of the data is stored on the rover's internal drive while some of it is transmitted through Wi-Fi back to the base station.

3.3.5. Communication

The communication subsystem handles transmitting GNSS positioning and signal strength from the Forest Buddy and its base station. This allows the rover to communicate its location while it collects signal strength measurements that can be used to tell how healthy forested lands are. Additionally, the subsystem provides the user with a remote controller to manually operate Forest Buddy when it's needed. For the GNSS positioning and signal strength measurements the antenna must be able to receive L1 and L2 satellite signals while having reliable performance in an outdoor environment.

Three GNSS Antennas were compared based on frequency, gain, durability and cost. Table 3-1 lists the possible GNSS options and selection.

Table 3-8: GNSS Antenna Options

Product	Cost	Frequency Compatibility	LNA Gain	Environmental Protection	Weight
Requirements	≤$200 Per Antenna	Must Support GPS L1 and L2	≥20 dB	≥ IP52	≤350g
SparkFun GNSS Multi-Band L1/L2 Surveying Antenna – TNC(TOP106)[19]	$199.95 per antenna	L1: 1555 to 1615 MHZ L2: 1198 to 1278 MHZ	40 ± 2 dB	IP67	335g
SparkFun GNSS L1/L2 Muti-Band Magnetic Mount Antenna-5m(SMA)[20]	$109.95 per antenna	0.8 cm + 1 ppm	28 ± 3 dB	IP67	175g
Ardusimple Budget Survey Multiband GNSS Antenna[21]	$99.05 per antenna	0.8 cm + 1 ppm	40 dB average	IP66*	390g

The SparkFun GNSS Multi-Band L1/ L2 Surveying Antenna – TNC(TOP106) was chosen over other options because of its survey grade design and suitability for precise GNSS positioning. The TOP106 features a ground plane that reduces multipath interference caused by reflected satellite signals. This helps when operating in a forested terrain. The 5m(SMA) was not chosen because its LNA gain wasn't as strong as the TOP106. The ArduSimple antenna was not chosen because it exceeded the weight limit while it had similar LNA gain to the TOP106. The IP rating could also fluctuate depending on how the user handles the air vent in the screw hole. It should also be mentioned that the TOP106 was given by the team's faculty advisor, so the cost for two antennas was $0 total.



Fig. 3-11: SparkFun GNSS Multi-Band L1/L2 Surveying Antenna- TNC(TOP106) [19]

The TNC(TOP106) Antenna will be connected to a compatible GNSS receiver which will then connect to a telemetry radio. The Antennas GNSS position will be as precise as 0.01m when connected to the telemetry radios for the RTK connection. The higher LNA gain will help amplify the weak signal from the satellites overhead, and it also has a ground plane integrated into the antenna to help reduce multipath interference.

Table 3-9 shows the options for the GNSS receiver options, the requirements needed for Forest Buddy, and the specifications that best fit the needs of the rover.

Table 3-9: GNSS Receiver Options

Product	Cost	RTK Position Accuracy	Number of UART Interfaces	Cold-Start Time
Requirements	≤$200 per receiver board	≤ 1 cm + 1 ppm	≥2	≤ 30s
Sparkfun GPS-RTK-SMA Breakout- ZED-F9P(Qwiic)[22]	$259.95	1 cm + 1 ppm	2	25s
SparkFun Quadband GNSS RTK Breakout – LG290P[23]	$189.95	0.8 cm + 1 ppm	3	28s
Waveshare LGP290P Series Quad-Band GNSS Module[24]	$89.99	0.8 cm + 1 ppm	3	28 s

The SparkFun GPS-RTK-SMA Breakout- ZED-F9P was chosen because it follows the cost, RTK positional accuracy, number of UART interfaces, and cold start time. The team was given the ZED F9P by the faculty advisor, so the cost was $0 for both boards. The RTK positional accuracy is within 1 cm, the UART interface is 2 and the cold start time is twenty-five seconds. While this doesn't out-perform either the SparkFun Quadband or the Waveshare LGP290P, except for the cold start time, the zero cost alone eliminates the two boards. The team has successfully implemented and tested the SparkFun ZED- FP9 board.



Fig. 3-12: SparkFun GPS-RTK-SMA Breakout-ZED-F9P [22]

Figure 3-12 shows the GNSS receiver module that was selected for Forest Buddy's base station and rover. The SparkFun board provides centimeter level accuracy which satisfies the requirement of being within a 10 cm radius of the rover's position. The cold start time is twenty-five seconds which gets its original GNSS position in that time. There will be two boards, each connected to the TOP106 GNSS antennas. The boards are also connected to two telemetry radios which help send signal data and help with RTK correction between the base station and rover.

Table 3-10 shows the different options for telemetry radios, requirements needed and the specifications that best fit the needs for the rover.

Table 3-10: Telemetry Radio Options

Product	Cost	Operating Frequency (MHz)	Communication Range	Transmit Power
Requirements	≤$200	902 – 928 MHz	≥ 200m	≤1 W
HolyBro SiK Telemetry Radio V3[25]	$117.98	915 MHz	300-500 m	100 mW
HolyBro SiK Long Range – 1W[26]	$175.00	902-928 MHz	50 km	1 W
B-CUBE 3DR Radio Telemetry [27]	$87.00	915 MHz	≥300m	100 mW

The HolyBro SiK Telemetry Radio V3 was chosen because its compatible with the Pixhawk 4 and its communication range is 300-500 meters which is more than capable of handling our requirements. The radios operate at 915 MHz and have a RF transmit power of 100 mW. The HolyBro radios are compatible to the ZED F9P GNSS receiver boards and soldered to those boards already. The team was given two sets of these radios so there is no cost to the project.



Fig. 3-13: HolyBro SiK Telemetry Radio V3 [25]

There will be two telemetry radios connected to the ZED-F9P GNSS receivers. One will be at the base station while one is on the rover. The radios transmit RTCM correction data from the base station to the rover which improves the positional accuracy. They also can send signal measurements for attenuation analysis which allows the team to compare the satellite signal strengths between the rover and base station.

Table 3-11 shows the remote controller options for Forest Buddy, requirements needed for the rover, and the specifications that best fit the rovers needs.

Table 3-11: Remote Controller Options

Product	Cost	Operating Frequency	Channels	Compatible with Pixhawk 4?
Requirements	≤$150	2.4 GHz	≥ 6	Yes
FlySky FS-i6X + FS-iA6B[28]	$69.99	2.4 GHz	6	Yes
Sony DualShock 4(PS4)[29]	$64.99	2.4 GHZ	6 buttons, 2 joysticks, 1 D pad, 2 triggers	Must be connected through Bluetooth
RadioLink AT9S Pro + R9DS[30]	$157.97	2.4 GHz	10	Yes

The FlySky FS-i6X remote controller was selected because of its low cost, its compatible with the Pixhawk 4, and enough channels to help control the rover. It's less expensive than the RadioLink AT9S Pro. The Sony DualShock controller requires Bluetooth and more coding work to connect it to the Pixhawk 4 which is why it was not chosen. The FlySky operates at 2.4 GHz and gives the user six channels to operate in autonomous mode and operator mode while also setting waypoints for the rover to follow. It can also directly talk to the Pixhawk 4 using PPM.



Fig. 3-14: FlySky FS-i6X + FS-iA6B Remote Controller [28]

The remote controller will set the waypoints for the rover to follow. The rover travels through the forest using human tracking to follow the user through its first path. While the operator walks in front of the rover, they set waypoints for a safe path for Forest Buddy. The operator will be able to switch between autonomous mode and operator mode which allows for the user to control the rover when needed.

3.3.6. Subsystem 6 (Data Analysis)

The data analysis subsystem displays GNSS signal data on a website that is processed by the data processing subsystem in a manner where users can read and understand results with efficiency. It stores processed data from the rover and the base station in a local file, and if there is a strong enough internet connection, it sends the data to a cloud-based database where from there it is sent to a website that displays data as set in the design requirement specifications. The website shows the data as a heat map with 5 labeled attenuation levels and a legend, and as a line graph of attenuation over time, and the number of points shown matches the number of records in the database. The subsystem is entirely software based; the base station uploads the records to the database through a cellular modem. To stay within budget and the 60-minute operating time, the database and website also have no recurring cost, store at least one complete 60-minute run (up to 54,000 records at 15 Hz), and accept up to 15 records per second.

Three hosted database services were compared. Supabase is a hosted PostgreSQL database with a built-in REST interface and a Realtime service that pushes table changes to connected web pages [31]. InfluxDB Cloud Serverless is a database designed specifically for time-stamped sensor data [32]. Cloud Firestore is a document database from Google Firebase [33]. Table 3-12 compares the free plan of each option against the subsystem requirements.

Table 3-12: Cloud Database Options

Cloud Database	Cost	Storage	Write Rate	Data Retention	Live Updates	Idle Behavior
Supabase (PostgreSQL)	Free plan	500 MB	No write limit	Until deleted	Realtime push	Pauses after 1 week unused
InfluxDB Cloud Serverless	Free plan	Enough for run	About 17 kB/s	Deleted after 30 days	Website must keep checking	Always Available
Cloud Firestore	Free plan	1 GiB	20,000 writes per day	Until deleted	Live listeners	Always Available

Supabase is selected as the cloud database. It is the only option that meets every requirement on its free plan: it stores many 60-minute runs within its 500 MB limit [31], has no daily write quota that a single run can exceed, and keeps data until the team deletes it. Its table editor also shows the stored records and their values directly, which supports the Requirement 3 test of comparing the website with the database. InfluxDB is designed for this type of data and handles the expected load, but its free plan deletes data after 30 days [32], which removes test data before the end of the semester. Firestore is eliminated because one run exceeds its limit of 20,000 writes per day [33]. The drawback of Supabase, that an idle project pauses after one week [31], is handled by opening the project at least weekly during development. Data from each run is also exported to CSV after the run and kept as a backup.

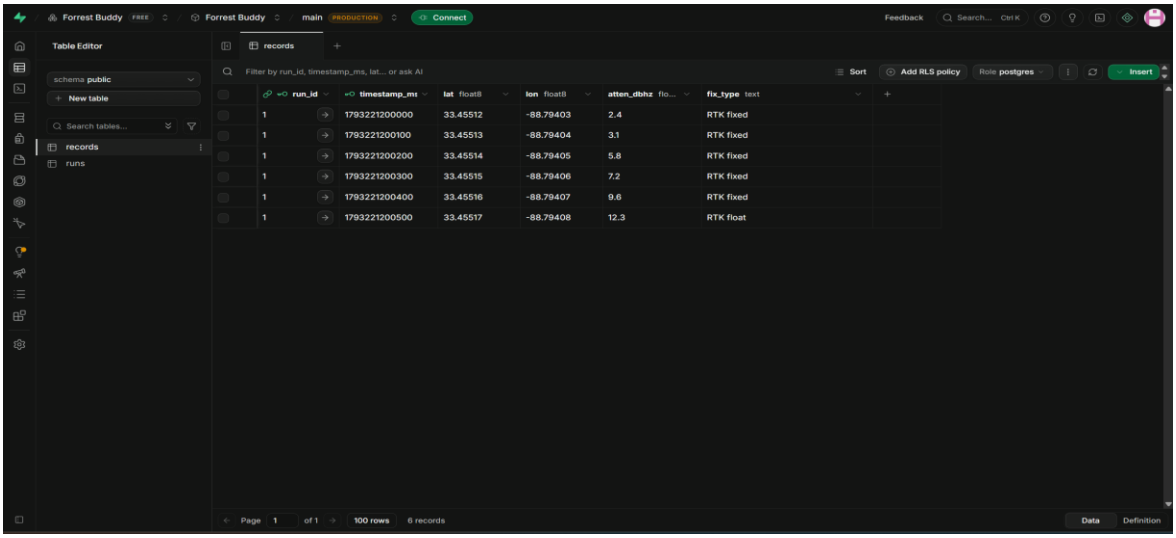


Figure 3-15: Supabase Project Table Editor

Figure 3-15 shows the Supabase table editor for the Forest Buddy project. Each row is one record from the rover, and the columns match the data format in Table 3-12. This view is also used in the Requirement 3 test to read the number of stored records and the stored attenuation values of the 3 selected points.

Three ways to build the website were compared. Grafana Cloud provides ready-made dashboard panels that connect to a PostgreSQL data source [34]. Streamlit Community Cloud hosts dashboards written entirely in Python [35]. The third option is a custom static web page that uses the Leaflet mapping library [36] and the Plotly.js charting library [37], reads data directly from Supabase, and is hosted for free on GitHub Pages [38]. Table 3-13 compares the options.

Table 3-13: Cloud Database Options

Cloud Database	Cost	Heat Map	Line Graph	Data Check	Availability	Build Effort
Grafana Cloud	Free plan	Limited legend control	Available	Needs extra setup	Always on	Low
Streamlit Cloud	Free plan	Needs add-ons	Available	Needs costum Code	Sleeps after 12 hours idle	Low
Custom Web page	Free plan	Full control	Available	Built into the page	Always on	Medium

The custom static web page is selected. It gives full control over the heat map levels and legend required by Requirement 1, and the point count and click-to-view values required by Requirement 3 are built directly into the page. Because the page is a set of static files, it has no server that can sleep or crash during the demonstration. Grafana is a strong backup option if development time runs short, since it connects to the same database without any changes. Streamlit is eliminated because the app sleeps after 12 hours without visitors [35] and must be woken before a demonstration.

Each record received from the Data Processing subsystem contains a timestamp, the rover position, and an attenuation value that is already calculated. The Data Analysis subsystem does not recalculate, average, or filter these values. Every record is drawn as one point on the heat map and one point on the line graph, so the number of points shown always equals the number of records stored, as Requirement 3 specifies. For the heat map, each point is assigned a level from 1 (lowest attenuation) to 5 (highest attenuation). The level ranges are set once field data shows the actual range of attenuation along the demonstration road. Figure 3-16 shows the flowchart of the website software.

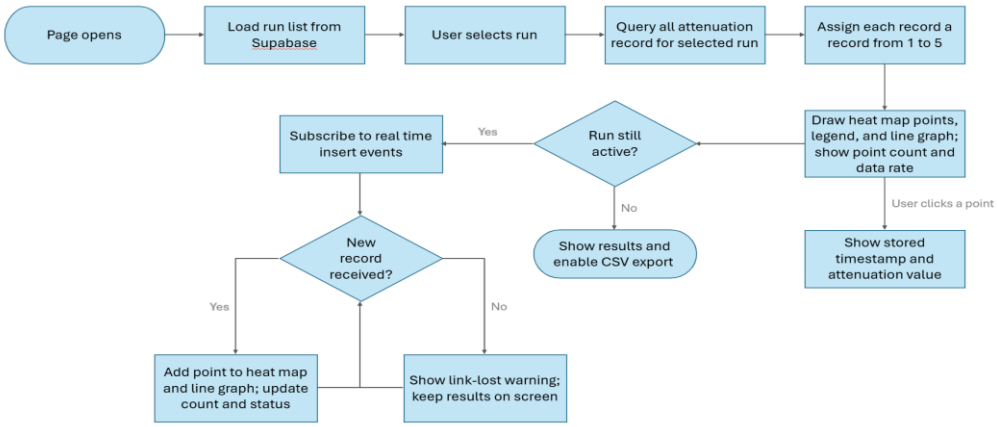
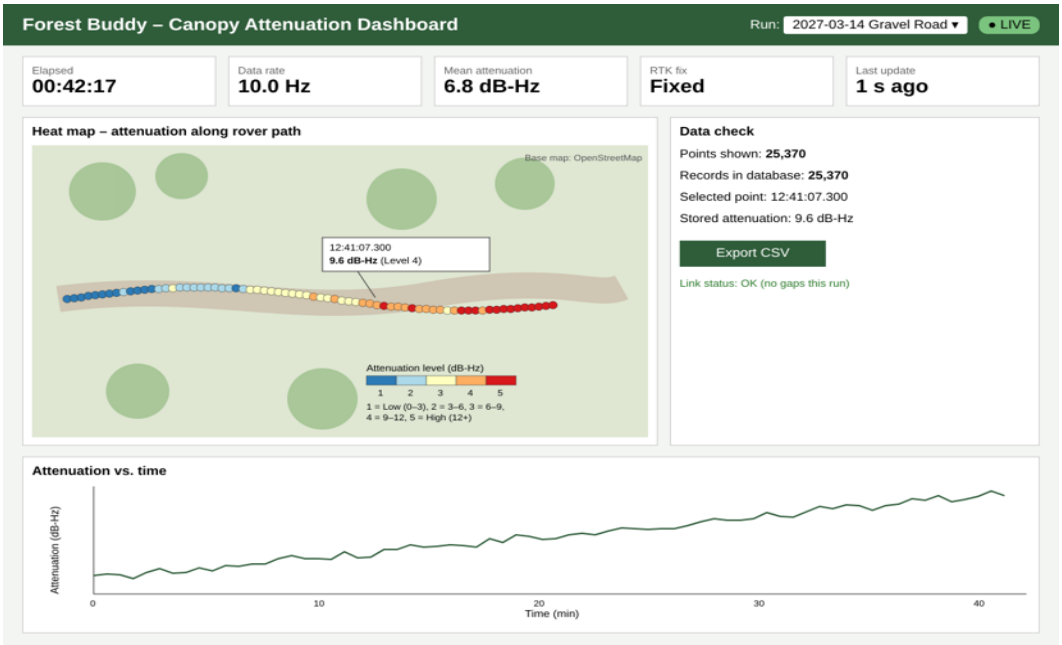


Figure 3-16: Data Analysis Website Flowchart

The above figure 3-16 shows how the website works. When the page opens, it loads the list of runs and the user selects one. The page then pulls every record for that run, assigns each record a level, and draws the heat map with its legend and the line graph of attenuation in dB-Hz against time. It also shows the point count next to the record count from the database, and the data rate in hertz, which is used for the Data Processing logging rate test. If the run is still active, the page listens for new records and adds each one as it arrives. If no data arrives for 30 seconds, a link-lost warning appears while the existing results stay on screen. Clicking any point shows its timestamp and stored attenuation value, which allows the 3-point check in Requirement 3.

Each record has a unique key made of the run and timestamp. If a record is sent twice after a dropped connection, the database overwrites it instead of storing a copy, which keeps the record count in Requirement 3 accurate. The database also allows the website to read while new records are being written, so the upload and the website never wait on each other.

Figure 3-17 shows the mockup of the Forest Buddy website.



(AI generated by Claude)

Figure 3-17: Data Analysis Website Mockup

The top row of the website shows elapsed time, data rate, mean attenuation, RTK fix status, and time since the last update. The heat map shows every record along the rover path, colored by its attenuation level, with a legend for the 5 levels. The data check panel shows the number of points on the page next to the number of records in the database, along with the timestamp and stored value of the selected point. The line graph at the bottom plots attenuation in dB-Hz against time with both axes labeled. A CSV export button downloads the data for the selected run.

Figure 3-18 shows how the software components of the subsystem connect.

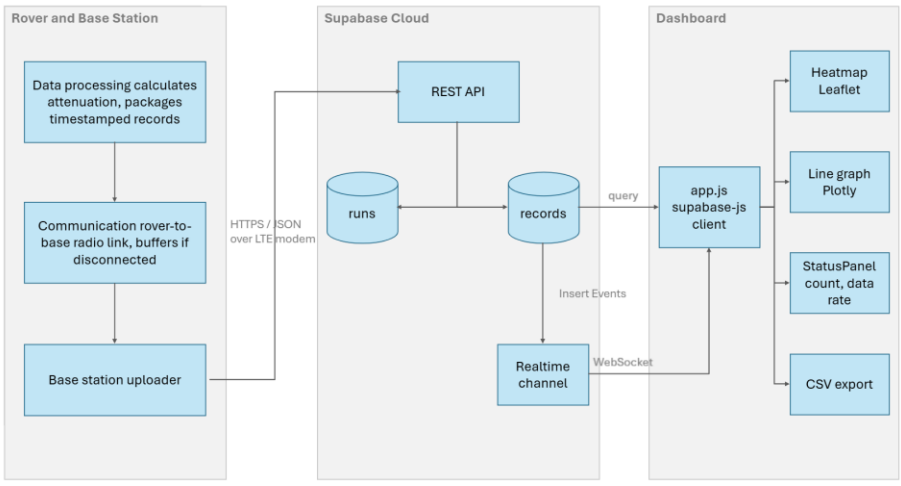


Figure 3-18: Shows the software components of the subsystem connect

The Data Processing subsystem packages the records, the Communication subsystem carries them to the base station, and the base station uploader sends them to Supabase, where they are stored in the runs and

records tables. On the website, app.js uses the Supabase JavaScript client to pull the records for the selected run and to receive new records as they arrive. It then passes them to the heat map, line graph, status panel, and CSV export components. Together, these pieces take the packaged attenuation data from the rover to a heat map and line graph that the team can view during and after each run.

3.4. Level 2 Prototype Design

The full prototype of Forest Buddy to be completed in Design II integrates all subsystems shown in Fig. 3-19 into a functional semi-autonomous rover and base station. The rover's onboard systems are controlled using the Jetson Orin Nano mini-PC, managing the environmental data, GNSS data, camera processing, sensor fusion and waypoint navigation, while the raspberry pi handles the data visualization at the base station. Power is regulated using a buck convert for lower power components. The 20A speed controller handles motor control with commands from the Jetson Orin Nano. When in initial operation, Forest Buddy follows a human operator through the forest, then renavigates through the same path from the set waypoints.

3.4.1 Level 2 Diagram

The design is implemented with the rover base station model, using a base station to store data and act as static GNSS baseline, while the rover antenna is connected to the mobile robot, with the telemetry radios communicating for RTK and transmission between the two. The rover features a Jetson Orin Nano as a microprocessor. For locomotion, the Rover's chassis is built from 4 2318 motors atop connected carbon fiber boards, controlled with 2 20A bidirectional speed controllers with four wheels. For navigation, the rover uses the BNO086 imu for orientation, the FHL-LD19 Lidar for mapping surroundings and detecting obstacles with a navigation algorithm consisting of obstacle avoidance and waypoint navigation. The camera vision subsystem consists of a C920 webcam to capture the environment alongside a tracking and detecting computer vision algorithm for tracking human operators. The signal processing subsystem uses a combined temperature, humidity, pressure, and gas sensor board in the BME688 for taking environmental data while using a Kalman filter to combine this data with GNSS information. The communication subsystem consists of the two GNSS TOP106 antennas, the two ZED-F9P receivers, and the HolyBro SiK telemetry radios. The communication subsystem handles the connection between the remote controller and rover, communicates the GNSS position data between the rover and base station for maintaining RTK position through the telemetry radios, and uploads packeted data from signal processing to base station. The data analysis subsystem consists of a Wi-Fi module for Wi-Fi based data transmission, a cloud database, and website for post processing, hosting, and data visualization. The sensor suite and jetson nano are enclosed in a plastic box affixed to the carbon fiber chassis. The two carbon fiber boards are covered in 3d printed filament, with the wires covered in waterproof connections and the edges sealed with silicon sealant.

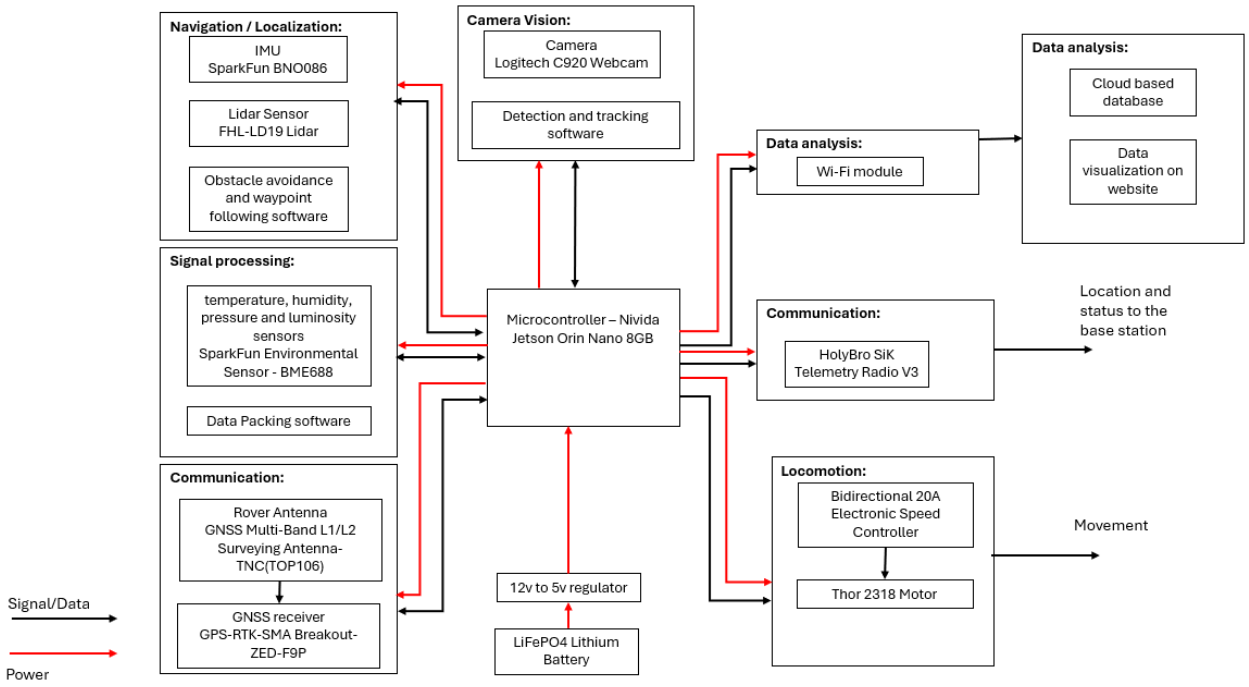


Fig. 3-19: Diagram for Forst Buddy (Level 2)

Figure 3-19 depicts a detailed overview of the rover design, featuring subsystem connections and software components of relevant subsystems.

The previous sections outline the design approach for Forest Buddy. All six subsystems must work together and become one singular rover. Due to the interconnectedness of Forest Buddy, if one subsystem fails then the rover itself will fail. The next sections show the testing environments and test run to ensure the subsystems work properly and can be easily implemented into one rover.

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